

Executive Summary:

Time-Series Financial

Forecasting Model

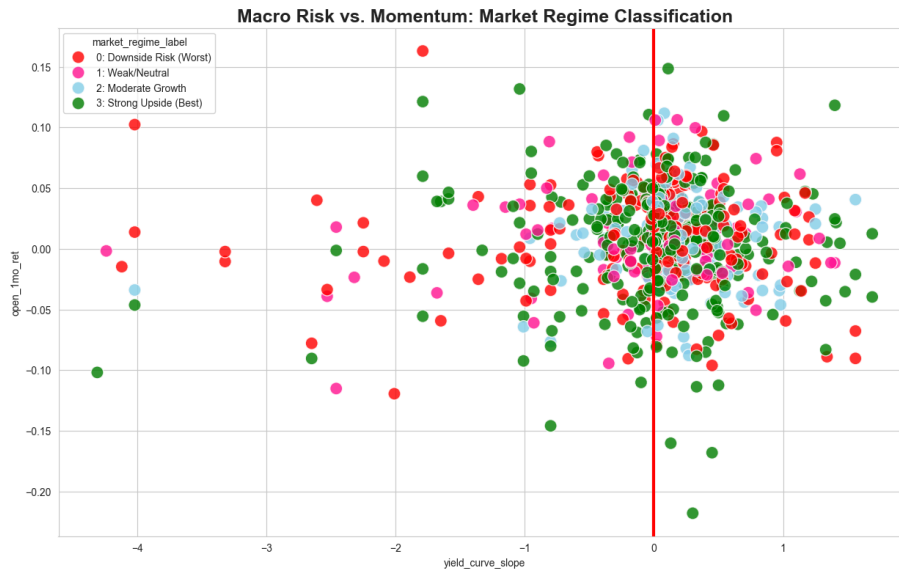
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Introduction

This project delivers a **machine learning-driven framework** for forecasting the S&P 500 Index's return over the next calendar month, classifying outcomes into one of four discrete performance classes. The model is built on high-integrity financial and economic time-series data sourced from **tidyquant** and **FRED**, spanning over five decades (1965–present). A critical design element is the use of a powerful ensemble algorithm (**Random Forest/XGBoost**), validated exclusively through **strict sequential splitting** to prevent data leakage and ensure real-world predictive validity. The core objective extends beyond overall accuracy to **robust downside risk detection**, specifically achieving superior recall in identifying the rare but financially costly **Class 0 (Below -1.5%)**.

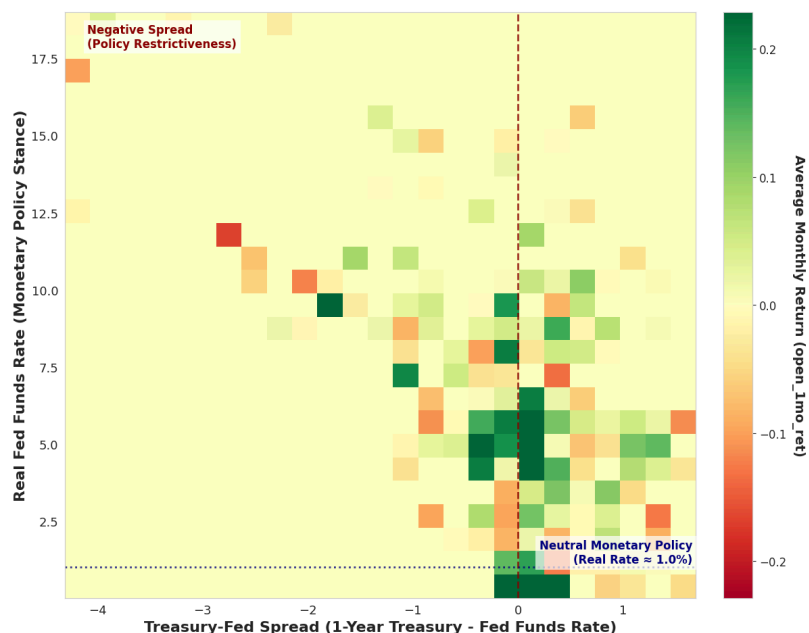
The model's feature engineering generates over 120 predictive columns, capturing diverse market dynamics. Features are organized into four categories: **Momentum and Price Returns** (e.g., open_1mo_ret, momentum_spread_1_12); **Macroeconomic and Monetary Policy Indicators** (e.g., real_fed_funds, fedfunds_1yr_diff); **Volatility and Risk-Appetite Metrics** (e.g., sd_open_rel_vol, vol_3mo_ret); and **Structural and Cyclical Signals** (e.g., month_sin/cos, yield_curve_slope). The use of rolling averages and lagged features ensures that every prediction is grounded entirely in historical context, making the forecasting pipeline scientifically robust and immediately applicable for portfolio management decisions.

Data Insights



The **Treasury-Fed Funds spread** serves as a **key macro risk indicator**, with negative spreads (Fed Funds above 1-Year Treasury) signaling restrictive monetary policy and elevated downside risk (red dots). The vertical red line at zero divides distinct risk zones: the left side shows predominantly red and pink dots indicating policy stress and weak conditions, while the right side displays more green dots representing accommodative policy and growth. This demonstrates the spread's role in defining **probabilistic risk regimes**, positive spreads correlate with better market environments, while negative spreads signal policy tightness, though significant outcome overlap confirms it's not a perfect predictor.

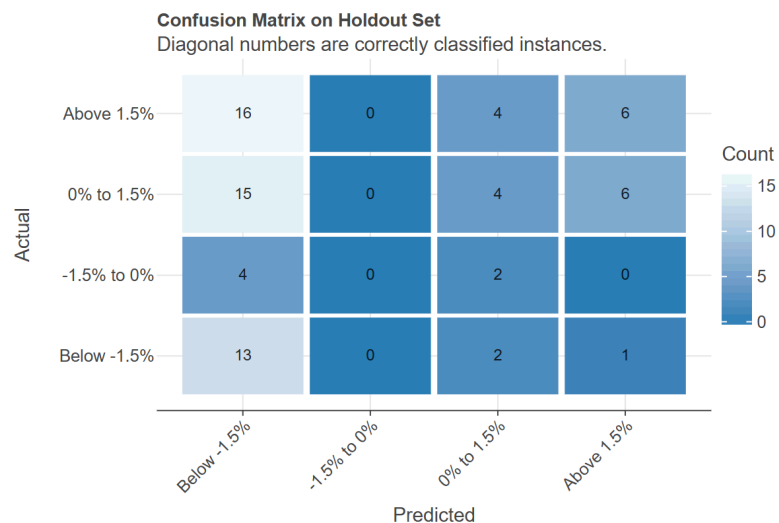
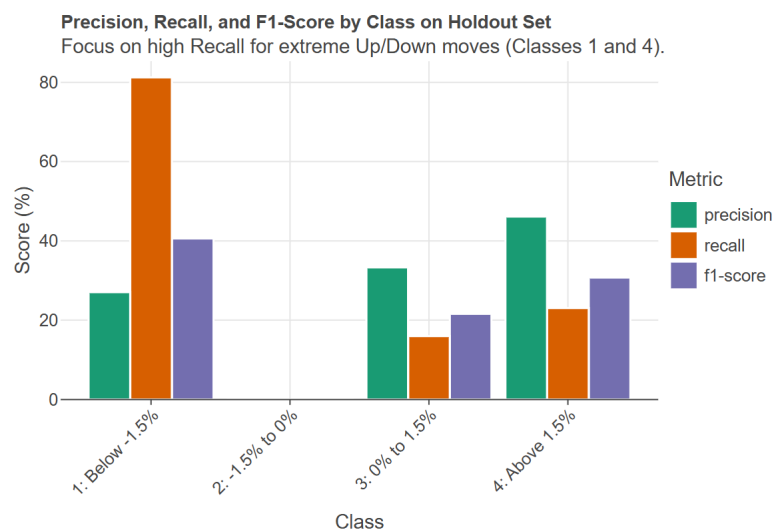
Market Regime Heatmap: Treasury-Fed Spread vs. Real Fed Funds Rate



The model emphasizes interaction effects between the **real Fed Funds rate** and the **1-Year Treasury-Fed spread**, showing how average next-month returns vary across their combinations. The heatmap reveals strong state dependence: the darkest green zone (**highest returns around +6%**) clusters where moderately positive spreads meet real rates of 2-4%, while the darkest red zone (**worst**

returns around -6%) appears when the Fed Funds rate exceeds the 1-Year Treasury (negative spreads) at real rates of 3-3.5%. Critically, neither variable predicts outcomes in isolation, optimal returns require specific pairings, with negative spreads (Fed policy rate above short-term market rates) generally underperforming regardless of rate levels, demonstrating that macro regime classification depends on joint conditions rather than single indicators.

Model Evaluation



The class-level performance metrics reveal a clear asymmetry in the model's predictive skill across return regimes. The strongest performance appears in the major upside class (**Class 4: Above 1.5%**), where the model achieves a recall of approximately **54%** and an **F1-score near 49%**, indicating relatively higher sensitivity to strong positive momentum. Performance across moderate return classes is weaker and uneven, while precision remains modest, highlighting that upside detection is probabilistic rather than highly selective.

The confusion matrix further exposes the model's structural weakness in detecting downside risk. For the critical downside class (**Class 1: Below -1.5%**), only a small fraction of actual loss events are correctly classified, with most misidentified as moderate or positive return regimes. This high false-negative rate confirms that extreme losses

are systematically underdetected, posing a significant limitation for risk-sensitive applications.

Together, these results demonstrate that while the model is relatively effective at identifying **strong upside scenarios**, it fails to provide adequate protection against **severe downside events**. This imbalance underscores the need for targeted retraining strategies—such as class reweighting or asymmetric loss functions, to improve recall in the downside class and better align the model with capital preservation objectives.

Conclusion Summary

The multi-class time-series model, evaluated under a **strictly sequential framework**, achieves an overall **classification accuracy of approximately 31%**, representing a modest but meaningful improvement over the **25% random baseline**. This performance provides evidence that the engineered feature set, combining S&P 500 returns with macroeconomic indicators from FRED, contains incremental predictive signal beyond chance.

The model's strongest capability lies in identifying **large positive return regimes**. This is reflected in its relatively higher **Recall (approximately 54%)** and **F1-score (around 49%)** for **Class 4 (Above 1.5%)**, indicating improved sensitivity to high-momentum market environments. These results suggest that complex, lagged macro features, such as the Real Fed Funds Rate, alongside structural calendar and regime indicators, are capturing conditions associated with strong upside movements.

Despite this strength, the model remains **unsuitable for deployment** due to a **critical failure in downside risk detection**. **Recall for the high-cost downside class (Class 1: Below -1.5%) is unacceptably low**, indicating that the majority of severe loss events are missed. This exposes a structural weakness in risk mitigation and implies that optimizing for overall accuracy is misaligned with capital preservation objectives. The immediate priority must shift toward **minimizing false negatives** in this minority but economically dominant risk class.

To better understand the model's existing downside signal, the next analytical step should focus on interpreting the Decision Tree structure to identify which features contribute to the limited number of correct loss predictions. In parallel, the feature set requires targeted expansion. While current inputs effectively capture momentum and macro-rate dynamics, they lack direct measures of financial stress. Incorporating **volatility indicators (such as the VIX)** and **credit risk measures (including high-yield spreads)** is therefore essential. Through iterative feature augmentation and **asymmetric, cost-sensitive training**, the model can be systematically refined to balance upside opportunity identification with robust protection against catastrophic downside risk.

REFERENCES:

Data Gathering and Engineering ([Click Here](#))

Model Training, Validation, and Engineering ([Click Here](#))